keylog.js: An Open Source Pedagogical Tool for DH and Data Studies

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Questions about the ethics, privacy, and accessibility of technologies and algorithms are central to digital humanities and data studies (DIgnazio and Klein 2020; Jobin, Ienca, and Vayena 2019). The software keylog.js provides a pedagogical tool for studying one such technology through a process of hands-on critical making and doing (Péron, Moody, and Karasic 2022; Ratto 2011).

Keystroke logging, or keylogging, involves recording the time and identity of the keys a user presses on a keyboard (Deckert 2021; Pawan 2023). Frequently run without a user's consent, malicious applications of keylogging include stealing private login credentials and other personal data (Rai et al. 2022). Through the analysis of relatively short samples, it is possible to use keylogging information as a method for tracking and surveillance across websites without a user's knowledge or consent (Elelegwu et al. 2024; Nyang, Mohaisen, and Kang 2014; Stefan, Shu, and Yao 2012). However, not all applications of keylogging are pernicious. Keylogging data is an essential and comparatively unobtrusive source for academic applications in linguistics, psychology, cognitive science, medicine, and human-computer interaction (Ballier et al. 2024; Berninger 2012; Braund et al. 2023; Conijn et al. 2021; Sullivan and Lindgren 2006). The power and simplicity of keylogging techniques make them an excellent topic for building awareness of the ethical and social aspects of existing and emerging technologies.

keylog.js is a minimal javascript-based tool that provides privacy-focused, client-side keylogging software served through a static website (Arnold 2024). No programming knowledge is needed to use the software. The static website and all of the code are hosted on an open GitHub repository. A screenshot of the tool is shown in Figure 1. The software records every keystroke and mouse click inside of the text entry box. However, importantly, the data never leaves a user's local computer and never includes anything done outside of the text entry box. Users can save a dataset of their keylogs prduced by the software formatted as a CSV file that any standard spreadsheet software or programming language can open. An example of the data produced is shown in Figure 2.

The keylog.js software can be applied in a classroom setting as follows. First, students are directed to the static website that hosts the tool. Students are asked to read the disclaimer on the website and given the chance to raise any questions or concerns. Next, students proceed to enter text in the tool in response to a prompt. The goal is to select a prompt that does not appear to contain personal information, such as: "describe the objects on the right-side of the room" or "provide directions for a visitor to go from our classroom to the main library." After responding to the prompt, students are asked to save and open their data in a spreadsheet program. Once everyone is finished, we have a class discussion centered around questions that ask us to reflect on the kinds of information contained in the output and the potential consequences of a third party generating a similar dataset using our data on another website. A set of sample questions are given in the Appendix.

The activity aims to demonstrate the ease at which an invasive picture of one's activity can be built using a short script running in the browser. Keylogging is particularly interesting for a class discussion because it raises questions regarding legal issues, state-sponsored surveillance, and concerns about capitalistic logic while offering potential positive outcomes such as a medical diagnostic tool (Shukla, Shukla, and Patel 2023; Singh et al. 2023). This range of issues requires students to think deeply and critically about the role of technology and AI in their lives, which requires careful consideration of positive and negative outcomes and how to balance them (Benatti, Gooding, and Sillence 2021). Allowing students to generate and interrogate their keylog data through the privacy-focused keylog.js software adds an important extra dimension to the discussion while helping empower students to ask better questions about the various closed-source software and platforms that they interact with daily.

The above activity can and has been done with students without programming or technical prerequisites. Additionally, keylog.js offers applications in courses that have a computational component. The resulting dataset is quite complex and can be used to study a variety of important concepts, such as window functions and date-time objects (Wickham, Çetinkaya-Rundel, and Grolemund 2023). Threading a dataset used in a privacy- and ethics-centered activity throughout the semester reinforces that these topics are a fundamental and pervasive aspect of data science praxis that should always be at the forefront of

any computational application (Bile Hassan et al. 2021; Saltz, Dewar, and Heckman 2018).

Finally, the software's code follows best practices for open-source software (Barker et al. 2022; Del Pico, Gelpi, and Capella-Gutiérrez 2024; Wilkinson et al. 2016). It is released under a permissive license (MIT), contains a contributor code of conduct, guidelines for contributors, and was built with no external dependences or build tools to minimize security vulnerabilities.

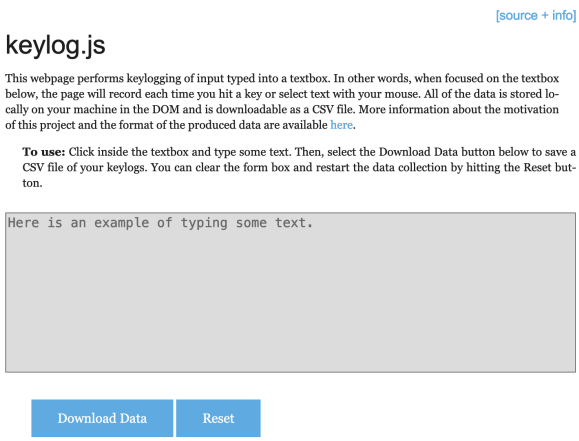

Figure 1. Screenshot of the keylog.js web interface, shown after typing some text and downloading the results.

time	type	key	key_code	alt_key	ctrl_key	meta_key	shift_key	is_repeat	range_start	range_end
14929	click	NA	NA	FALSE	FALSE	FALSE	FALSE	FALSE	0	0
16505	down	Shift	ShiftLeft	FALSE	FALSE	FALSE	TRUE	FALSE	0	0
16601	down	H	KeyH	FALSE	FALSE	FALSE	TRUE	FALSE	0	0
16605	input	H	insertText	FALSE	FALSE	FALSE	FALSE	FALSE	1	1
16629	up	Shift	ShiftLeft	FALSE	FALSE	FALSE	FALSE	FALSE	1	1
16673	up	h	KeyH	FALSE	FALSE	FALSE	FALSE	FALSE	1	1
16696	down	e	KeyE	FALSE	FALSE	FALSE	FALSE	FALSE	1	1
16700	input	e	insertText	FALSE	FALSE	FALSE	FALSE	FALSE	2	2
16763	down	r	KeyR	FALSE	FALSE	FALSE	FALSE	FALSE	2	2
16766	input	r	insertText	FALSE	FALSE	FALSE	FALSE	FALSE	3	3
16783	up	e	KeyE	FALSE	FALSE	FALSE	FALSE	FALSE	3	3
16867	down	e	KeyE	FALSE	FALSE	FALSE	FALSE	FALSE	3	3
16870	input	e	insertText	FALSE	FALSE	FALSE	FALSE	FALSE	4	4
16923	up	r	KeyR	FALSE	FALSE	FALSE	FALSE	FALSE	4	4
16986	down	NA	Space	FALSE	FALSE	FALSE	FALSE	FALSE	4	4
16989	input	NA	insertText	FALSE	FALSE	FALSE	FALSE	FALSE	5	5
17007	up	e	KeyE	FALSE	FALSE	FALSE	FALSE	FALSE	5	5
17094	down	i	KeyI	FALSE	FALSE	FALSE	FALSE	FALSE	5	5
17097	input	i	insertText	FALSE	FALSE	FALSE	FALSE	FALSE	6	6
17111	up	NA	Space	FALSE	FALSE	FALSE	FALSE	FALSE	6	6

Figure 2. The first few rows of the output from the keylog.js tool as shown in Excel. The rows in the image show the process of typing the first six characters—"Here i"—from Figure 1.

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- Figure 3. Screenshot of the project's public source code on GitHub. The site can be visted at https://github.com/taylor-arnold/keylog.js.

Below are example discussion questions used when applying the **keylog.js** software in digital humanities, data studies, and data science courses. After these relatively generic questions, we proceed to connect these topics to the specific readings that we have covered in the course.

- Would you find it invasive to publically share the data produced by the tool of your response? If so, what aspects would you be concerned about?
- Did you ever use the backspace in your response? Look and see how that was shown in the data, noting that the erased text still exists in the results. Does this change your response to the question above?
- If we collected all of this data across the class, we could sort everyone along dimensions such as slowest typer, most erasures, most long pauses, etc. Does this change any of your responses above?
- What are some potential applications of keylogging? Try to think of some that might be malicious and some that might be benevolent.
- What guidelines and/or guardrails would you recommend to someone applying keylogging in a research application? Or, should we avoid the technique altogether?

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